

Protein Structures Alignment Algorithms by Parametric Searching with Trigonometric Series

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Abstract

Proteins are an important class of biological macromolecules present in all biological organisms. Protein structure is essential for correct function because it allows molecular recognition. Thus protein structure provides the opportunity to recognize homology that is undetectable by sequence comparison, and it represents a powerful means of discovering functions, yielding direct insight into the molecular mechanisms. Currently, there are several techniques available in attempting to find the optimal alignment of shared structural motifs between two proteins.

In this paper, we propose algorithms and develop tools for pairwise alignment of protein structures. Methods of locating suitable isometric transformations of one structure and aligning it to the other structure are addressed. The proposed methods allow sequence gaps of any length, reversal of chain direction, and free topological connectivity of atom sequences. One of the crucial steps of the algorithm is finding a good isometric transformation, which leads to the best atom-pairing alignment between two proteins. Based upon the periodical property of the parametric settings, we propose parametric searching strategies by approximations with power series and trigonometric series. We show the effectiveness of the proposed refinement methods by a set of experiments, which improve several previous results.

Keywords: structural proteomics, structure alignments and comparisons, approximation, trigonometric series, power series

1 Introduction

Protein structures play critical roles in vital biological functions [11]. With more than 40,000 protein structures determined by the advances in X-ray crystallography and NMR spectroscopy to date, molecular biologists these days proceed in the direction of analyzing and classifying these protein structures in order to discover the structural relationships with protein functions [8].

Detection of proteins with a similar fold can suggest a common ancestor, and often a similar function [7]. Comparison of 3D structures makes it possible to establish distant relationships, even between protein families distinct in terms of sequence comparison alone. This is why structural alignment of proteins increases our understanding of more distant evolutionary relationships [5, 14]. The link between structural classification and sequence families enables us to study functions of various folds, or whole proteins [16].

There have been several methods proposed to compare protein structures and measure the degree of structural similarity based on alignment of secondary structure elements as well as alignment of intra and inter-molecular atomic distances. The basic ideas are rapid identification of pair alignments of secondary structure elements, clustering them into groups, and scoring the best substructure alignment. For examples, the VAST system [6] is based on continuous distribution of domains in the fold space. The FSSP/DALI system [13] provides two levels of description – a coarse-grained one and one with a fine-grained resolution. The method, CATH, provides the complete PDB fold classification by domains and links to other sources of information. The two methods, CE and LGscore2 [23] focus on the local geometry rather than global features such as orientation of secondary structures and overall topology (as in the case of VAST or DALI). VAST has been used to compare all known PDB domains to each other. The results of this computation are included in NCBI's Molecular Modelling Database

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at <http://www.ncbi.nlm.nih.gov/Structure/-VAST/vast.html>.

Note that there must be an atom-pairing scheme before one can do the structure alignment computation. The first atom of the first selection is compared to the first atom of the second selection, fifth to fifth, and so on. Incorporating with ideas of bipartite matching and 3-parameter isometric transformation, Lin *et al.* [15, 22] proposed methods of using parametric searching strategies with adaptive controls, and demonstrated that more accurate and similar protein structure pairings are possible comparing to previous known results like VAST [6] or CE [23].

One of the crucial steps of these algorithms is finding a good isometric transformation, which leads to the best atom-pairing alignment between two proteins. In this paper, we propose algorithms of for efficiently locating more suitable isometric transformations of one structure and aligning it to the other structure. Based upon the periodical property of the parametric settings, we propose parametric searching strategies by approximations with power series and trigonometric series. We show the effectiveness of the proposed parametric searching strategies by a set of experiments, which leads to better alignments of structure pairing in general.

2 Background and Terminology

Consider the point of north-pole $\mathbf{n} = (0, 0, 1)$ on the unit sphere. After the rotation, R , say $\mathbf{n}$ is rotated to another point $\mathbf{p} = (x, y, z)$; i.e., $\mathbf{p} = R\mathbf{n}$. Let α denote the angle $\angle \mathbf{nOp}$. Note that α determines the z -coordinate of $\mathbf{p}$. To determine x -coordinate and y -coordinate of $\mathbf{p}$, the point is rotated around the z -axis for the angle β on the unit sphere. Note that there are infinitely numbers of rotation that transform $\mathbf{n}$ to $\mathbf{p}$. The particular rotation R can be decided by rotating all other points around the vector $\mathbf{p}$ by the angle γ . It is not hard to verified that, in such a way, *any* rigid rotation transformation can be parameterized by the three-tuple (α, β, γ) . Thus, we call a vector $\mathbf{p} = (x, y, z)$ on the surface of the unit sphere a *probe*. Note that the *movement* of each probe is started from the north-pole $(0, 0, 1)$ to other points in the sphere. The position of $\mathbf{p}$ is decided by the parameters (α, β) , and exact rotation is fixed by the self-rotation angle γ .

The main idea of our algorithm for finding a suitable matching between two sets of points before utilizing the RMSD procedure to fine-tune the final result is by searching the suitable (parametric) probe. After that, we use the minimum bipartite matching algorithm to find the best matching

between two sets to decide the best matching for the RMSD procedure. Let $P' = T \circ P$, and Q being translated to Q' such that the mass center of Q' is located at the origin. We construct a weighed graph $G = (V, E)$ with V being labelled with points of P' and Q' , and each (p, q) in E being weighted with the squared Euclidean (3D) distance; i.e., $w(p, q) = \|p, q\|^2$. We then solve the weighted *minimum bipartite matching* problem [10] to obtain the best matching of P' and Q' . By the matched pairing, we perturb and refine the final alignment to obtain a possible lower *rmsd*.

By incorporating with the two key concepts, parameterized-rotation as well as bipartite matching, the main algorithm can compare paired protein structures once given a *reasonable good* initial setting of the 3 parameters. The proposed generic parametric searching algorithm is showed in Figure 1. Since the best parametric settings can be very difficult to locate, our previous methods concentrate on using randomized perturbation method in searching sufficiently large number of parametric probes over the parameter spaces and let each probe searching its own proximity in a randomized greedy manner.

Here in this paper we demonstrate that the underlying corresponding *rmsd* values associated with the parametric sets are related to each other in periodical and *continuous* manner; thus seeking reasonable approximation of the underlying *rmsd* values distributions is possible by some suitable mathematical models, especially by trigonometric series.

2.1 Root mean squared deviation

The smallest *root mean squared deviation* (*rmsd*) is a least-squares fitting method for two sequences of points [13]. The idea is to align atom vectors of the two given (molecular) structures, and use the common least averaged squared errors as a measurement of differences between these two (paired) sequences. Formally, let $P = \langle p_1, \dots, p_n \rangle$ and $Q = \langle q_1, \dots, q_n \rangle$ be two sequences of points. We assume that P is translated so that its centroid $(\frac{1}{n} \sum_{k=1}^n p_k)$ is at the origin. We also assume that Q is translated in the same way. For each point or vector x , let $(x)_i (i = 1, 2, 3)$ denote the i -th (X, Y, Z) coordinate value of x , and $\|x\|$ denote the length of x . Let

$$\text{RMSD}(P, Q, R, \mathbf{a}) = \sqrt{\frac{1}{n} \sum_{k=1}^n \|Rp_k + \mathbf{a} - q_k\|^2}, \quad (1)$$

where R is a rotation matrix and $\mathbf{a}$ is a translation vector. Then, the *rmsd* value $d(P, Q)$ between P and Q is defined by $d(P, Q) = \min_{R, \mathbf{a}} d(P, Q, R, \mathbf{a})$.

Although complicated as it might appear, the optimal rotation matrix and translation vector can be found simultaneously in $O(n)$ time. Schwartz [21] showed that $d(P, Q, R, \mathbf{a})$ is minimized when $\mathbf{a} = 0$ and

$$R = (A^t A)^{\frac{1}{2}} A^{-1}, \quad (2)$$

where the matrix $A = (A_{ij})$ $i, j = 1, 2, 3$ is given by

$$A_{ij} = \sum_{k=1}^n (p_k)_i (q_k)_j, \quad (3)$$

where $A^{\frac{1}{2}} = B$ means $BB = A$, and $\mathbf{o}$ denotes the zero vector. Thus, $d(P, Q)$, R and $\mathbf{a}$ can be computed in $O(n)$ time [18].

We adopt Martin's ProFit package (standing for protein fitting system) [17] to calculate the *rmsd* between C- α atoms of paired protein backbones. ProFit has many features including flexible specification of fitting zones and atoms, calculation of RMS over different zones or atoms, RMS-by-residue calculation. Fitting was performed using the McLachlan algorithm [18].

2.2 Isometric Rotation Transformation

According to Euler's rotation theorem [9], any rotation about the origin point can be described by using three angles. The rotation is determined by 3 consecutive rotations with 3 *Euler angles* (α, β, γ) . The first rotation is done by the angle α around the z -axis, the second is done by the angle β around the x -axis, and the third rotation is done by the angle γ around the z -axis.

As a result, we reduce the problem of finding a good rotation matrix to the new problem of finding a good 3-parameter. The rotation matrix is thus characterized by just adjusting the 3 uniformly distributed parameters.

The 3-parameter Rotation Matrix

Similar to Euler's rotation transformation, our 3-parameter method (α, β, γ) can be summarized as the following:

- **Rotation around z -axis:**

Given a unit vector $\mathbf{p} = (x, y, z)^T$, $\mathbf{p}$ is transformed into $\mathbf{p}'$ by a rotation around the z -axis by angle α . That is, let

$$\mathbf{p}' = \begin{pmatrix} x_\alpha \\ y_\alpha \\ z_\alpha \end{pmatrix} = \begin{bmatrix} \cos \alpha\pi & \sin \alpha\pi & 0 \\ -\sin \alpha\pi & \cos \alpha\pi & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \mathbf{p}$$

- **Rotation around x -axis:**

The vector, $\mathbf{p}' = (x_\alpha, y_\alpha, z_\alpha)^T$, is transformed into the probe $\mathbf{p}''$ by a rotation around the x -axis by angle β . That is, let

$$\mathbf{p}'' = \begin{pmatrix} x_\beta \\ y_\beta \\ z_\beta \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \beta\pi & -\sin \beta\pi \\ 0 & \sin \beta\pi & \cos \beta\pi \end{bmatrix} \cdot \mathbf{p}'$$

then we will get new coordinate of $(x_\beta, y_\beta, z_\beta)^T$.

- **Rotation around the probe $\mathbf{p}''$:**

The last rotation matrix, R_γ , do the body rotation around the probe $\mathbf{p}''$ by angle γ ; see [12] for related discussions about the transformation. That is, let

$$(x, y, z) = (x_\beta, y_\beta, z_\beta)^T.$$

$$c = \cos \gamma\pi, s = \sin \gamma\pi, h = 1 - c.$$

$$R_\gamma = \begin{bmatrix} c + x^2h & xyh - zs & xzh + ys \\ xyh + zs & c + y^2h & yzh - xs \\ xzh - ys & yzh + xs & c + z^2h \end{bmatrix}$$

Minimum Bipartite Matching

There are several proposed algorithms for the minimum bipartite matching problem; sometimes it is also referred as the *assignment problem*. Here we adopted the Munkres [19, 4, 3, 20] algorithm. The public available implementation is written with Perl language. To improve the efficiency of computation, we implement the Munkres algorithm and write hundreds lines of C Codes, and the produced codes are strictly verified by comparing the results with the public Perl package. The bipartite matching procedure is incorporated with the parameterized rotation to set up various orientation settings.

Randomized Perturbation

Note that the methodology of the current system is generally a randomized algorithm and a variable through PERTURB algorithm. The performance of PERTURB algorithm is depended on various range of θ_{max} . It can help us to have a random parameter around the assigned θ . The perturb algorithm is displayed in figure 2.

3 Parametric Estimation

In order to obtain the structural alignment that is free from the topological connectivity of atom sequences, the proposed generic parametric searching algorithm shown in Figure 1 relies on a *reasonable good* initial setting of the 3 parameters.

STRUC-PE($P, Q, \alpha_I, \beta_I, \gamma_I, \mathbf{p}$) $\triangleright$ Structure Alignment with Parametric Estimation.

Input: Two set of 3D coordinates of points $P = \{p_1, p_2, \dots, p_n\}$ and $Q = \{q_1, q_2, \dots, q_m\}$; $n < m$.
 The α_I , β_I and γ_I are real numbers that are between -1 to 1.
 $\triangleright$ These inputs control the initial position of 3 parameters box and affect the explored area.
 $\triangleright$ $\mathbf{p}$ is the vector $(x, y, z)^T$, explained in section 2.3.

Output: $(s, \alpha, \beta, \gamma)$ is a sufficiently low RMSD s and (α, β, γ) .
 $\triangleright$ (α, β, γ) is the best position of 3 parameters box.

Global : $T, t, \theta, \text{deg}[i]$
 The threshold T is a integer number.
 $\triangleright$ T is total trial number of perturbation.
 $\triangleright$ t is the count that records the number of perturbation carried out so far.
 $\triangleright$ θ is an temporary array to save α, β, γ .
 $\triangleright$ $\text{deg}[i]$ is the degree of the estimation function for $\theta[i]$.

- 1 $(\alpha, \beta, \gamma) \leftarrow (\alpha_s, \beta_s, \gamma_s) \leftarrow (\alpha_I, \beta_I, \gamma_I)$.
- 2 $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\alpha, \beta, \gamma, \mathbf{p}))$. $\triangleright Q'$ is a temp array of atoms set of protein.
- 3 $L \leftarrow \text{MBM}(P, Q')$; $(R, \mathbf{a}) \leftarrow \text{MS-FIT}(L, P, Q')$; $s \leftarrow \text{RMSD}(P, Q', R, \mathbf{a})$.
- 4 $t \leftarrow 0$. $\triangleright$ Initializing the counter.
- 5 **while** $t \leq T$ **do**
- 6 **for** $i \leftarrow 1$ to 3 **do**
- 7 $(\theta[1], \theta[2], \theta[3]) \leftarrow (\alpha, \beta, \gamma)$. $\triangleright$ Reset parameters $\theta[i]s'$.
- 8 **if** $\text{deg}[i] = 5$ **then** $t \leftarrow t+4$ **else** $t \leftarrow t+1$.
- 9 $\theta[i] \leftarrow \text{ESTIMATE}(P, Q, \theta[1], \theta[2], \theta[3], i, \text{deg}[i])$.
- 10 $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\theta[1], \theta[2], \theta[3], \mathbf{p}))$. $\triangleright Q'$ is a temporary array of atoms set of protein.
- 11 $L \leftarrow \text{MBM}(P, Q')$; $(R, \mathbf{a}) \leftarrow \text{MS-FIT}(L, P, Q')$; $s' \leftarrow \text{RMSD}(P, Q', R, \mathbf{a})$.
- 12 **if** $s' \leq s$ **then** $s \leftarrow s'$; $(\alpha, \beta, \gamma) \leftarrow (\alpha_s, \beta_s, \gamma_s) \leftarrow (\theta[1], \theta[2], \theta[3])$; **for** $j \leftarrow 1$ to 3 **do** $\text{deg}[j] \leftarrow 5$.
- 13 **else** $\text{deg}[i] \leftarrow \text{deg}[i] + 2$.
- 14 **return** $(s, \alpha_s, \beta_s, \gamma_s)$.

MBM(P, Q) returns the minimum bipartite matching of two point sets P and Q .
ESTIMATE($P, Q, \theta[1], \theta[2], \theta[3], i, \text{deg}[i]$) estimates a better parameter, $\theta[i]$, hopefully leading to a lower *rmsd*.

TRANS(A, R).
Input: A is an array of 3D points with size n .
 R is the rotation matrix.
Output: An array of 3D points, B .

- 1 **for** $i \leftarrow 1$ to n **do**
- 2 $B[i] \leftarrow R \cdot A[i]$ $\triangleright B$ is the array containing the transformed n points.
- 3 **return** B .

Figure 1: Aligning two sets of atoms with low *rmsd* by pairing points according to the minimum bipartite matching measurement

EST-POWER($P, Q, \theta[1], \theta[2], \theta[3], k, deg$) $\triangleright$ Estimate the parameter with lowest *rmsd*.
Input: The ($P, Q, \theta[1], \theta[2], \theta[3], k, deg$) $\triangleright$ Parameters are explained in Figure 1.
Output: (θ^*)
Global : θ_{max} $\triangleright$ Ranged 0 to 1, controls the parameters perturbation.

```

1  for  $i \leftarrow 1$  to  $deg$  do
2       $\theta_{temp} \leftarrow \text{PERTURB}(\theta[k], \theta_{max})$ .
3       $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\theta[1], \theta[2], \theta[3], \mathbf{p}))$ .
4       $L \leftarrow \text{MBM}(P, Q') ; (R, \mathbf{a}) \leftarrow \text{MS-FIT}(L, P, Q')$ ;
5       $s[i] \leftarrow \text{RMSD}(P, Q', R, \mathbf{a})$ .  $\triangleright s$  is an array to save the rmsd.
6       $\text{Matrix}[i][1] \leftarrow 1$ .  $\triangleright \text{Matrix}$  is an array to save the estimated function.
7      for  $j \leftarrow 2$  to  $deg$  do  $\text{Matrix}[i][k] = \theta_{temp}^{j-1}$ .
8       $C \leftarrow \text{GAUSSELM}(\text{Matrix}, s)$ .  $\triangleright$  The estimated function is  $f(\theta) = C[1] + C[2]\theta + C[3]\theta^2 + \dots$ 
9       $\theta' \leftarrow \text{EST-LOWEST}(C)$ .  $\triangleright$  Find the minimum value of  $f(\theta)$ .
10 return ( $\theta^*$ ).
```

PERTURB(θ, θ_{max}).

Input: θ is a real number between -1 to 1. θ_{max} is the Perturbation range of θ .

$\triangleright$ This input control the present position of 3 parameters box and affect the explore area.

Output: a real number (θ').

$\triangleright$ This output is the new position of the parameter box.

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1   $\Delta\theta \leftarrow \text{RAND}(-\theta_{max}, \theta_{max}) ; \theta' \leftarrow \text{ROUND}(\theta + \Delta\theta)$ .
2  return ( $\theta'$ ).
```

RAND(a, b) is a random function returning a real number uniformly distributed between a and b .

$$\text{ROUND}(\theta) = \begin{cases} 2 + \theta & \text{if } \theta \leq -1 \\ -2 + \theta & \text{if } \theta \geq 1 \\ \theta & \text{otherwise} \end{cases}$$

Figure 2: The algorithm of power series estimation method and the perturb algorithm.

Since the best parametric settings can be very difficult to locate, our previous methods concentrate on using randomized perturbation method in searching sufficiently large number of parametric probes over the parameter spaces and let each probe searching its own proximity in a randomized greedy manner.

However, since the corresponding *rmsd* values and parametric sets are related to each other in a periodical and (relatively) continuous manner, it is possible to seek for a reasonable approximation of the associated *rmsd* values distributions by some suitable mathematical models.

Using brute force manner to find all the possible parameters would cost too much time and computer resources to find the best solution, leading to the lowest rmsd. The idea is to find reasonable good parameters under the limited time and resources constraints. In the following we introduce the power (polynomial) series and trigonometric series to approximate the underlying associated *rmsd* values distribution corresponding to each parameter $\theta = \alpha, \beta, \gamma$. The experimental data will show which kinds of mathematical models better suits for the estimation function.

Gaussian Elimination

Gaussian Elimination [1] can be used to determine the solutions of a linear equations. When we use the power series method or the trigonometric series method to estimate the approximate curve, we have to find a series of unknown coefficients. It can be done by the Gaussian elimination to solve a linear equation in k variables. After we test and verify the perl package performing the Gaussian elimination, we incorporate the package into our system to solve the linear equations in our power series method and trigonometric series method.

3.1 Approximation with Power Series

Multinomial Series

In order to come up with a good approximation mathematical model for the underlying *rmsd* value distribution, our first choice was a estimated curve given by the multinomial series. Since the considered three parameters could be correlative to each others, we considered a multinomial series raised to the second power:

$$\begin{aligned} f(\alpha, \beta, \gamma) = & C_1 + C_2\alpha + C_3\beta + C_4\gamma \\ & + C_5\alpha\beta + C_6\alpha\gamma + C_7\beta\gamma \\ & + C_8\alpha^2 + C_9\beta^2 + C_{10}\gamma^2 \end{aligned} \quad (4)$$

where $f(\alpha, \beta, \gamma)$ denote the estimated value of *rmsd* under the 3-parameter (α, β, γ) . We test the function and find some sets of parameters. Unfortunately, after the model is tested against the real data in our database, we found the proposed model does not match with the underlying real *rmsd* distribution. One of the main reason of the failed attempts is the intertwined (α, β, γ) parameters yielding a confusing, and most of the times inconsistent, estimation model. One way to overcome the situation is by considering each parameter one at a time. So, instead of considering the 3 parameters all together, we adopt the strategy of tackling each direction one at a time and adjusting these parameters one by one. These parameters are treated as if they are independent of each other. Thus, for each parameter α, β, γ , we consider the following function:

$$f(\theta) = C_1 + C_2\theta + C_3\theta^2, \quad (5)$$

The equation can be easily generalized to the k th power, $k = 1, 2, 3, \dots$. We have the following estimation function for α, β, γ :

$$f(\theta) = C_1 + C_2\theta + C_3\theta^2 + \dots + C_k\theta^{k-1}, \quad (6)$$

Using the particular equation as the estimation function, the power series estimation algorithm is illustrated in Figure 2.

In order to evaluate the goodness of the estimated power series function over the underlying real *rmsd* distribution. The power series estimation method is tested against the PDB database with 2,000 random proteins pairs, and the tested result is summarized in Table 1. Let k be the power of estimation function. Note that in order to obtain a good guess of a particular parameter, say α , the k -th power estimation function is decided (by Gaussian elimination) from previous k arbitrary random guesses. Thus, if the conjectured θ^* of Figure 2 leads to an *rmsd*-value that is relatively small comparing to those *rmsd*'s given by the k random guesses, then we can conclude such an estimation function is a good one. Let the *rmsd*-value determined by the conjectured θ^* be m^* . The value is to be tested against k previous *rmsd*'s $m_1, m_2, \dots, m_k$. The term b stands for the number of m_i 's that is no smaller than m^* . We use *PR*-value and *t*-score to evaluate the effectiveness of our approximate strategy. The *PR*-value represent the percentage of opponents that is *beaten* by the estimations. That is,

$$PR\text{-value} = \frac{b}{k},$$

Furthermore, let $\mu(\sigma)$ be the mean (standard deviation) of $\{m_1, m_2, \dots, m_k\}$. The equation of *t*-score is given by

$$t\text{-score} = \frac{m^* - \mu}{\sigma},$$

Unfortunately, as we can see from the tested results shown in Figure 4, the power series estimation method for various degree k does not suggest any direction for the parameter leading to low *rmsd*. It is expected that any random guess shall obtain a *PR*-value about 50%. We conclude that the power series estimation method is no better than the random guess. One of the possible explanation is that, the Taylor series serves a good approximation for the continuous function only on the *proximity* of a narrow range. Since these k probes are randomly scattered upon the range $[-1, 1]$ with the number k relatively small, the power series approximation is just not good enough to capture the shape of *rmsd* distribution.

k	b (<i>PR</i> -value)	t -score	times of first rank
5	2.50 (50.02%)	-0.59	339 (16.9%)
7	3.50 (50.08%)	-0.58	241 (12.1%)
9	4.49 (49.91%)	-0.41	228 (11.4%)
11	5.50 (50.00%)	-0.43	180 (9.0%)

Table 1: The result of $k = 5, 7, 9, 11$. We show the rank, *PR*-value, *t*-score and the times of first rank by power series. The *PR*-value is about 50%. It is no better than random guess.

3.2 Approximation with Trigonometric Series

The idea of using trigonometric series as means of approximating the underlying *rmsd* distribution comes from a closer examination of Figure 4. The figure illustrates a typical representative of a protein pair that is randomly picked from the PDB database with random perturbed orientation. Each angle (parameter) of (α, β, γ) is exhaustively probed by the unit step as small as 0.005. The shown figure confirms our conjecture that the parametric *rmsd* distribution function in each direction is periodical and (relatively) continuous. Usually such curve is best approximated by a trigonometric series. It is only reasonable that trigonometric series shall be tested against the same protein pairs sample. Treating each parameter as an independent factor, the trigonometric series approximation function with degree k is listed as following:

$$\begin{aligned} f(\theta) = & C_1 + C_2 \cos \pi\theta + C_3 \sin \pi\theta \\ & + C_4 \cos 2\pi\theta + C_5 \sin 2\pi\theta \\ & + C_6 \cos 3\pi\theta + C_7 \sin 3\pi\theta + \dots \\ & + C_{k-1} \cos \frac{k-1}{2}\pi\theta + C_k \sin \frac{k-1}{2}\pi\theta \end{aligned} \quad (7)$$

where the $f(\theta)$ is the *rmsd*-value we try to approximate by one of the three parameters, (α, β, γ) , with the other two parameters fixed. We use the

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EST-TRIGO( $P, Q, \theta[1], \theta[2], \theta[3], k, deg$ )     $\triangleright$  Estimate the parameter with lowest rmsd
Input: The ( $P, Q, \theta[1], \theta[2], \theta[3], k, deg$ )     $\triangleright$  Parameters are explained in figure 1.
Output: ( $\theta^*$ )
Global :  $\theta_{max}$      $\triangleright$  Ranged 0 to 1, controls the parameters perturbation.
1  for  $i \leftarrow 1$  to  $deg$  do
2       $\theta_{temp} \leftarrow \text{PERTURB}(\theta[k], \theta_{max})$ .
3       $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\theta[1], \theta[2], \theta[3], \mathbf{p}))$ .
4       $L \leftarrow \text{MBM}(P, Q') ; (R, \mathbf{a}) \leftarrow \text{MS-FIT}(L, P, Q') ;$ 
5       $s[i] \leftarrow \text{RMSD}(P, Q', R, \mathbf{a})$ .     $\triangleright s$  is an array to save the rmsd.
6       $Matrix[i][1] \leftarrow 1$ .     $\triangleright Matrix$  is an array to save the estimated function.
8      for  $j \leftarrow 1$  to  $\frac{deg-1}{2}$  do
9           $Matrix[i][2j] \leftarrow \cos(j\pi\theta_{temp})$ 
10          $Matrix[i][2j+1] \leftarrow \sin(j\pi\theta_{temp})$ ;
11          $C \leftarrow \text{GAUSSELIM}(Matrix, s)$ .     $\triangleright$  The estimated function is  $f(\theta) = C[1] + C[2] \cos \theta + C[3] \sin \theta + \dots$ 
12          $\theta^* \leftarrow \text{EST-LOWEST}(C)$ .     $\triangleright$  Find the minimum value of  $f(\theta)$ 
13 return ( $\theta^*$ ).

```

Figure 3: The algorithm of trigonometric series estimation method.

function to test $k = 2$. We can find a estimated parameter and then we use the parameter to execute our program. By this equation we can adjust the three parameters one by one and increase the degree of the approximation function. Our algorithm is shown in Figure 3. We also draw a real figure about different values of β . It is shown in Figure 4. The estimated curve is created by trigonometric series compared with the real curve in α , β and γ . There are five estimative points when $k = 5$. Four of all are randomly picked, and another is 0. The range of random estimative points is -1 to 1. We can get the close curve in α and γ . But there are more peaks in the figure of β , we do not get a good graph. After we observe the figure of β , we find there are four peaks in its curve. We try $k = 11$ to execute the same graph, and then we have the following result in Figure 5. We indeed find a similar curve. We can know that the curve is close to trigonometric series. Therefore, we choose trigonometric series estimation method for the main part of this paper.

We use this algorithm to test some samples, and we have some results shown in Table 2. The trigonometric series estimation method for various degree of freedom k show the good result for parameter leading to low *rmsd*. We can get a *PR*-value about more than 63%, we conclude that the trigonometric series estimation method is better than the random guess.

4 Experimental Results

In this section, we show the experimental method and result. First, we introduce the data set of our experiment and how we choose the sample data in our experiment. We show how the

protein pairings data is collected from the real data. In the second part, we analyze the improvement ratios of our method tested upon the given data. Finally, we analyze the experimental execution time of our programs.

4.1 Data Set

We choose the PDB database [2] as our experimental source; the RCSB PDB is the main provider of PDB. We randomly pick 200 protein structures in the PDB database as our experimental subjects by uniform sampling. For each chosen protein structure, we randomly choose 30 structure alignment lists on the database of VAST as the tested targets. Totally, there are 6,000 protein pairings tested by the experiment. We use the term, P_A , to stand for one of the 200 randomly picked protein structures, and we use P_B to stand for one of the 30 neighbors of each P_A . We have 6,000 pairings for our population. We randomly choose 200 pairings from the population for our experiment. The sample distribution is showed in Figure 6(a). We use the 200 samples to carry following experiments and find out good parameters of the our programs.

k	b (<i>PR</i> -value)	t -score	times of first rank
5	3.75 (75.14%)	-2.42	950 (47.5%)
7	4.96 (70.83%)	-2.43	934 (36.7%)
9	5.92 (65.80%)	-1.18	578 (28.9%)
11	6.95 (63.20%)	-0.89	486 (24.3%)

Table 2: The result of $k = 5, 7, 9, 11$. We show the rank, *PR*-value, t -score and the times of first rank by trigonometric series. The *PR*-value is more than 63%. The trigonometric series estimation method is better than the random guess.

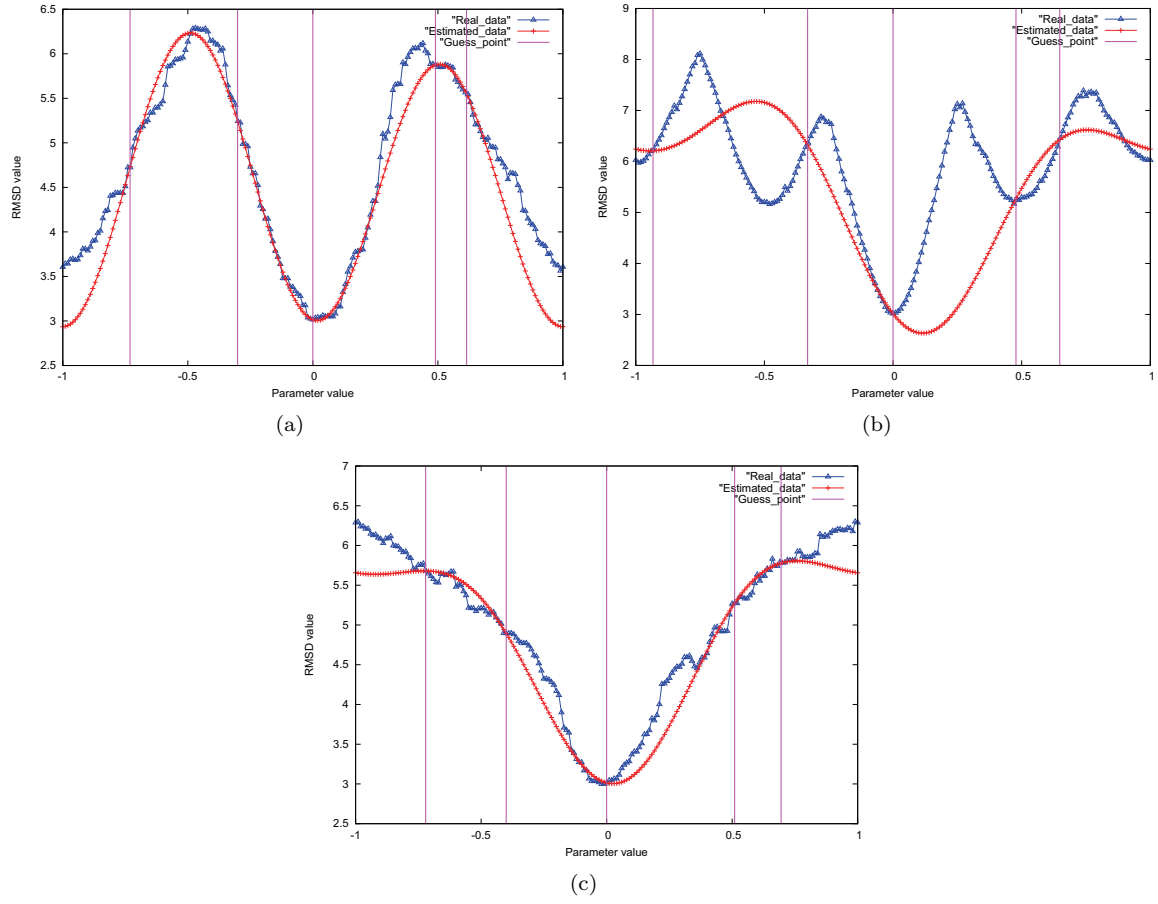


Figure 4: The estimated curve is created by trigonometric series compared with the real curve in α , β and γ . There are five estimative points when $k = 5$. Four of all points are randomly picked, and another one is 0. The range of random estimative points is -1 to 1. We can get the close curve in α and γ . But there are more peaks in the figure of β , we do not get a good graph.

The number of C- α atoms of P_A	The number of protein pairings	The average of improved ratio (%)	The standard deviation of improved ratio (%)
40<	39	17.50	12.70
80	47	15.33	8.71
120	42	14.22	10.20
160	23	15.46	7.504
200	15	20.37	10.90
240	13	15.93	10.34
280	14	11.56	11.39
300>	7	21.63	11.37
Total	200	15.89	10.17

Table 3: The result is to execute the algorithm of trigonometric series with initial alignment of VAST and compare with the VAST. The distribution of our experimental data containing the number of C- α atoms of P_A , the number of protein pairings, the average of improved ratio and the standard deviation of improved ratio

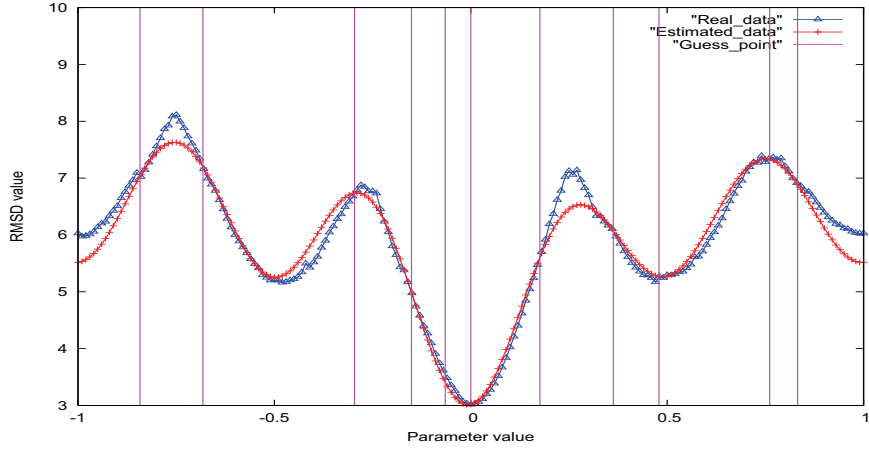
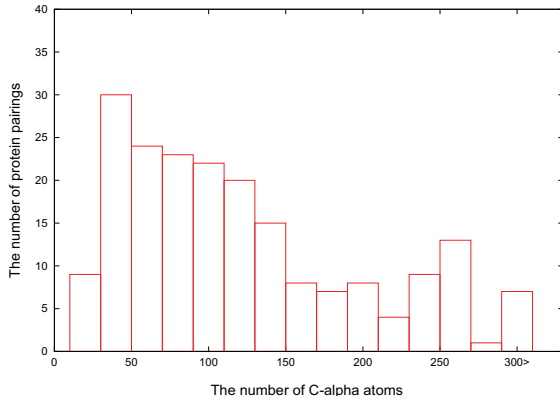
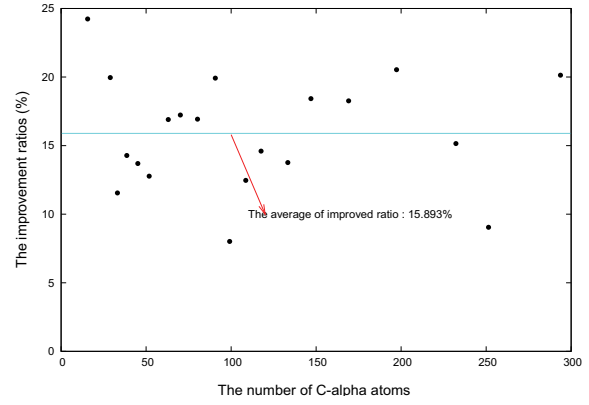


Figure 5: The estimated curve is created by trigonometric series compared with the real curve in β . We estimate the curve in $k = 11$, and there are 10 random estimative points, and the other one is 0. The range of random estimative points is -1 to 1. We indeed find a similar curve.



(a)



(b)

Figure 6: (a) The distribution of the 200 randomly picked protein structure pairings. (b) The improvement ratios opposite to the *rmsd* of the VAST alignment. The improvement ratios are mostly distributed from 9% to 20%. The average is 15.9%.

4.2 STRUC-PE with Trigonometric Estimation

Testing upon the 200 protein pairings, the experiment adopts the trigonometric series as the estimation method to improve parameters by the STRUC-PE algorithm. We first measures each sample pairings by calculating its original alignment's *rmsd* value. The measurement is confirmed and double checked with the *rmsd* by the *PROFIT* package kits [17].

The experimented results show that almost each of 200 tested samples is improved, comparing to the original VAST pairings. The samples can be partitioned into 20 clusters with each cluster containing 10 protein pairs in average. Figure 6(b) shows the relation between the number of C- α atoms of P_A and the improvement ratios contrast to the given VAST alignment. The formula of the

improvement ratio is defined by

$$\rho = \frac{A - B}{A},$$

where A is the original *rmsd* value by the VAST alignment, while B is improved(smaller) *rmsd* value by our structure alignment method. The improvement ratios of our samples are mostly distributed from 9% to 20%. As a total, the average of the improvement ratios is 15.9%.

For the mapping figures in Figure 4 and Figure 5, if the degree of trigonometric series is large enough (k is greater than or equal to the number of the peaks), the estimated curve of trigonometric series shown in can draw near the distribution of original curve. If k is strictly larger than the number of the peaks, such as Figure 4(c), it also have a reasonable well estimated curve by trigonometric series. It is interesting to note that Figure 4(b)

and Figure 5 are both tested against the same protein pair by probing on the β parameter. The only difference is that the approximation degree of Figure 4(b) is $k = 5$, while in Figure 5 we have $k = 11$. It is clearly seen that the approximation degree $k = 5$ is not sufficient to approximate the real data distribution, which possesses 4 peaks in the diagram.

5 Conclusion and Future Work

Bioinformatics has become an essential tool not only for basic research but also for applied research in biotechnology and biomedical sciences. Currently, the field is under going an enormous expansion, as witnessed by the dramatic increase in the number of related bioinformatic literatures.

In this paper, we use Struc-PE with trigonometric estimation to improve the *rmsd* value between a protein structure pair by finding a better alignment list. We use a set of experiment to test the parameters. We apply the tested parameters to run the experiment results. Our method can improve the alignment computed by the VAST by an averaged improvement ratios about 15.9%. By this method we can have better (lower *rmsd*) results and decrease the times to run minimum bipartite matching (MBM). The main compute for our method is to execute MBM. Therefore, the Struc-PE with trigonometric estimation indeed improves existed known system like the VAST.

Furthermore, since the structure comparison problem, like many scientific computation/simulation problem, is very time-consuming under cases of large structures and large number of paired structures, it is desirable to implement the system under massive parallel machines cluster, e.g., the grid-environment, to increase the throughput of the system.

There is an important direction of research along the topic of protein structure alignment. The problem of *local structure alignment* is to find the functional (or active) part of a given query protein; these active parts imply a substructure similarity between two proteins. The problem of identification of the similar substructure of a protein pair will indeed be a challenge.

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